Plant growth fields from non-rigid Coherent Point Drift

Registering day t onto day $t{+}1$: displacement field and $\det J$

Pheno4D: 7 maize and 7 tomato plants

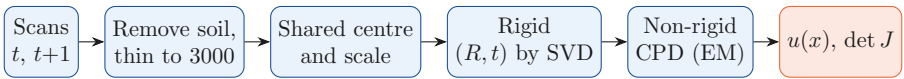
Key findings

- **Registration works.** Median Chamfer distance over the 14 plants drops from 13.6 to 2.1 (scan units), a median reduction of 81%.
- **13 of 14 fits are clean:** no folding ($\det J > 0$ at every point). For these, the median $\det J$ lies between 0.96 and 1.45 (median of medians 1.25).
- **One fit fails: T02** (Chamfer 6.5, 2.7% of $\det J \leq 0$). All 14 plants are run with rigid pre-alignment; it matters only for M06, which folds without it (90% of $\det J \leq 0$).
- **Caveat 1.** Large displacements are mostly leaf motion (drooping, swinging), so $\det J$ mixes growth with bending.
- **Caveat 2.** No ground truth on real data, and only one labelled consecutive pair (0324→0325) per plant.
- **Read** $\det J$ **qualitatively** (where tissue expands) and trust it only where the fit is clean.

Goal and data

- **Question:** where, between two consecutive days, does a plant grow?
- **Method:** register the day-t point cloud onto day $t{+}1$ with non-rigid CPD, take the smooth displacement field u, and compute the local volume change $\det(I + \partial u/\partial x)$.
- **Data:** Pheno4D (Schunck et al., 2021). It contains 7 maize and 7 tomato plants, so the result is 14 plots, not the 20 originally asked for.
- **Pair used:** the only labelled consecutive-day pair, 0324→0325. Soil is removed with label 0; each scan is thinned to 3000 points.
- **Per plant:** overlay, registered cloud, displacement field, $\det J$.

Pipeline



Shared centre and scale matter: normalising each scan separately would rescale the plant and erase the growth signal. Both scans use one centre c and one scale s.

Notation

$Y = \{y_m\}_{m=1}^{M}$	source points (day t)	$G_{ij} = e^{-\|y_i - y_j\|^2/2\beta^2}$	kernel matrix
$X = \{x_n\}_{n=1}^{N}$	target points (day t+1)	$W \in \mathbb{R}^{M \times D}$	coefficients, $T(Y) = Y + GW$
$D = 3$	dimension	P_{mn}	posterior: x_n came from y_m
$T(y)$	transformation of a point	$P1,\ P^\top 1$	row / column sums of P
$w \in [0, 1)$	outlier weight	$N_P = \mathbf{1}^\top P \mathbf{1}$	total matched mass
σ^2	Gaussian variance	$u(x)$	displacement field
λ	coherence (stiffness) weight	$J = I + \partial u/\partial x$	Jacobian of $x \mapsto x + u(x)$

Intuition: soft matching, moved together

No hard correspondences. Each source point is a Gaussian blob. Every target point is explained by all blobs, weighted by how close they are: a soft assignment P_{mn}.

Coherence. Points are not allowed to move independently. The penalty $\mathrm{tr}(W^\top GW)$ forces neighbours to move similarly, so the result is a smooth field rather than a pile of unrelated shifts.

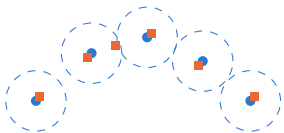
Annealing. σ^2 starts large (global, coarse match) and shrinks with the fit.

Three knobs

- w fraction of target points treated as unmatched (new tissue, noise). Used: 0.2.
- β distance over which points move together. Larger is smoother. Used: 1.2.
- λ weight of the coherence penalty. Larger is stiffer. Used: 30.

Units are normalised (shared scale). Early defaults $\beta = 0.3, \lambda = 3$ gave noisy $\det J$ with up to 17% negative values.

CPD model: a moving Gaussian mixture



dashed: Gaussians around $T(y_m)$
squares: target points x_n

The extra square between blobs is an outlier: the uniform term (weight w) explains it.

Mixture density of a target point (component 0 = outlier):

$$p(x_n) = \frac{w}{N} + \frac{1-w}{M} \sum_{m=1}^{M} \mathcal{N}\big(x_n \mid T(y_m), \sigma^2 I\big)$$

Motion model (non-rigid):

$$T(Y) = Y + GW$$

Penalised negative log-likelihood to minimise:

$$E = -\sum_{n=1}^{N} \log p(x_n) + \frac{\lambda}{2} \operatorname{tr}\big(W^\top G W\big)$$

Derivation 1: why EM

The sum inside the logarithm has no closed-form minimiser. Introduce a hidden label m_n (which component produced x_n) and alternate:

1. **E-step:** compute the posterior $P_{mn} = P(m \mid x_n)$ using the current (T, σ^2).
2. **M-step:** minimise the expected complete-data cost over (W, σ^2) with P fixed.

Each cycle cannot increase E. The expected complete-data cost is

$$Q(W, \sigma^2) = \frac{1}{2\sigma^2} \sum_{m,n} P_{mn} \|x_n - T(y_m)\|^2 + \frac{N_P D}{2} \log \sigma^2 + \frac{\lambda}{2} \mathrm{tr}(W^\top G W)$$

- First term: a weighted least-squares residual, each pair weighted by its match probability.
- Second term: from the Gaussian normaliser $(2\pi\sigma^2)^{-D/2}$ per matched pair, summed: $N_P = \sum_{m,n} P_{mn}$.

Derivation 2: the E-step (Bayes' rule)

Priors: $P(m) = \frac{1-w}{M}$ for $m = 1..M$, $P(0) = w$. Likelihoods: $p(x \mid m) = (2\pi\sigma^2)^{-D/2} e^{-\|x - T(y_m)\|^2/2\sigma^2}$, $p(x \mid 0) = \frac{1}{N}$.

$$P_{mn} = \frac{P(m)\, p(x_n \mid m)}{p(x_n)} = \frac{\frac{1-w}{M}(2\pi\sigma^2)^{-D/2} e^{-\|x_n - T(y_m)\|^2/2\sigma^2}}{\sum_{k=1}^{M} \frac{1-w}{M}(2\pi\sigma^2)^{-D/2} e^{-\|x_n - T(y_k)\|^2/2\sigma^2} + \frac{w}{N}}.$$

Multiply top and bottom by $(2\pi\sigma^2)^{D/2} \frac{M}{1-w}$:

$$P_{mn} = \frac{e^{-\|x_n - T(y_m)\|^2/2\sigma^2}}{\sum_{k=1}^{M} e^{-\|x_n - T(y_k)\|^2/2\sigma^2} + c}, \qquad c = (2\pi\sigma^2)^{D/2} \frac{w}{1-w} \frac{M}{N}$$

- Each column of P sums to less than 1; the missing mass is the probability that x_n is an outlier.
- Larger w increases c, so unmatched points absorb more weight.

Derivation 3: the M-step for W

Substitute $T(Y) = Y + GW$ and write $P1 = (\sum_n P_{mn})_m$. Setting $\partial Q / \partial W = 0$:

$$\frac{1}{\sigma^2}\Big[G\,\mathrm{diag}(P1)\,(Y + GW) - G\,PX\Big] + \lambda\, GW = 0$$

Multiply by σ^2 and factor G out of every term:

$$G\Big[\mathrm{diag}(P1)\,(Y + GW) - PX + \lambda\sigma^2 W\Big] = 0$$

G is positive definite, so the bracket is zero. Collect the terms in W:

$$\big(\mathrm{diag}(P1)\,G + \lambda\sigma^2 I\big)\,W = PX - \mathrm{diag}(P1)\,Y$$

- One $M \times M$ linear solve per iteration (the cost that limits us to about 3000 points).
- **Why this penalty:** among smooth fields it is the motion-coherence prior, and its minimiser is a sum of kernels on the source points, $u(x) = \sum_j G(x, y_j) w_j$. That closed form is what lets us differentiate u exactly.

Derivation 4: the σ^2 update

With T fixed, differentiate Q with respect to σ^2:

$$\frac{\partial Q}{\partial \sigma^2} = -\frac{1}{2\sigma^4} \sum_{m,n} P_{mn} \|x_n - T(y_m)\|^2 + \frac{N_P D}{2\sigma^2} = 0$$

$$\sigma^2 = \frac{1}{N_P \, D} \sum_{m,n} P_{mn} \, \|x_n - T(y_m)\|^2$$

In matrix form, $\sigma^2 = \frac{1}{N_P D} \left[\operatorname{tr}(X^\top \operatorname{diag}(P^\top 1) X) - 2 \operatorname{tr}((PX)^\top T) + \operatorname{tr}(T^\top \operatorname{diag}(P1)\, T) \right]$.

- It is the match-weighted mean squared residual per coordinate.
- Large σ^2 early means wide, forgiving Gaussians: a coarse global match. As the fit tightens, σ^2 shrinks and matching sharpens. This built-in annealing is what makes CPD robust to initialisation.
- Iterate E and M until σ^2 and E stop changing.

Rigid pre-alignment (used for all 14 plants)

Used when the whole plant has shifted or rotated between scans (no scale: plants grow, so scale is not a nuisance). Model $T(y) = Ry + t$. Minimise $\sum_{m,n} P_{mn} \|x_n - Ry_m - t\|^2$.

1. Match-weighted means: $\mu_x = \frac{1}{N_P} X^\top P^\top 1$, $\mu_y = \frac{1}{N_P} Y^\top P 1$. Centre: $\hat{X} = X - \mu_x$, $\hat{Y} = Y - \mu_y$.
2. Cross-covariance and SVD: $A = \hat{X}^\top P^\top \hat{Y} = U \Sigma V^\top$.

$$R = U \operatorname{diag}\big(1, 1, \det(UV^\top)\big) V^\top, \qquad t = \mu_x - R\mu_y$$

- The $\det(UV^\top)$ factor rules out reflections.
- A rotation has $\det R = 1$, so it does not change $\det J$.
- Effect here: identical to the default for 12 plants, it fixes M06 (90% $\rightarrow$ 0% folding), and it does not rescue T02.

Algorithm

Input: X (day $t{+}1$), Y (day t), $w = 0.2$, $\beta = 1.2$, $\lambda = 30$.

Prepare: remove soil, thin to 3000 points, centre by c and scale by s (shared by both scans). Rigid step: $Y \leftarrow RY + t$.

Init: $W = 0$, $\sigma^2 = \frac{1}{DMN} \sum_{m,n} \|x_n - y_m\|^2$, $G_{ij} = e^{-\|y_i - y_j\|^2 / 2\beta^2}$.

Repeat until σ^2 and E converge:

1. **E:** P_{mn} from the posterior formula.
2. **M:** solve $(\mathrm{diag}(P1)G + \lambda\sigma^2 I)W = PX - \mathrm{diag}(P1)Y$; set $T = Y + GW$; update σ^2.

Output: T (composed with the rigid step), W; field $u(x) = \sum_j G(x, y_j) w_j$ and $J(x)$ at any x (next slides).

From registration to a growth field

The fitted map is $\phi(x) = x + u(x)$ with

$$u(x) = \sum_{j=1}^{M} G(x, y_j)\, w_j \qquad (w_j = \text{row } j \text{ of } W).$$

Jacobian:

$$J(x) = \frac{\partial \phi}{\partial x} = I + \frac{\partial u}{\partial x}.$$

One kernel, $G(x, y_j) = e^{-\|x - y_j\|^2 / 2\beta^2}$, differentiates to

$$\frac{\partial G(x, y_j)}{\partial x_k} = -\frac{x_k - y_{jk}}{\beta^2}\, G(x, y_j)$$

so, entry by entry (the formula in the code):

$$J_{ik}(x) = \delta_{ik} - \frac{1}{\beta^2} \sum_{j=1}^{M} W_{ji}\, (x_k - y_{jk})\, G(x, y_j)$$

What $\det J$ measures

A small volume element dV at x becomes $\det J(x)\, dV$ at $\phi(x)$.

- $\det J > 1$: local expansion (growth).
- $\det J < 1$: local compression.
- $\det J \leq 0$: the map folds over itself, so the fit is not trustworthy there.
- Composition: $\det J_{\phi_2 \circ \phi_1} = \det J_2 \cdot \det J_1$, so $\log \det J$ adds over successive days.

Plants are thin surfaces. The field is only constrained along the surface, so a volume ratio is a rough proxy. The area change of a surface with unit normal n is

$$\frac{dA'}{dA} = \det J \left\| J^{-\top} n \right\|$$

(Nanson's formula). This is the better growth measure; it needs surface normals and is listed as a next step.

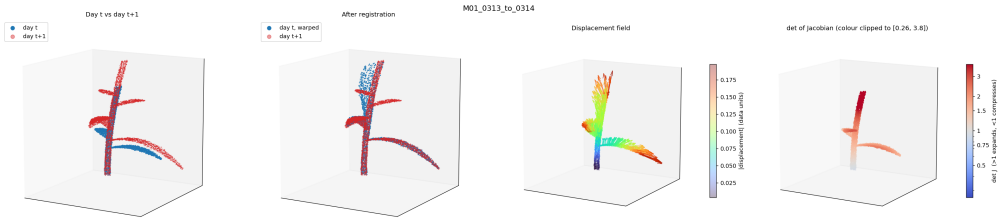
Validation 1: the Jacobian code

- Analytic J against central finite differences of ϕ: largest absolute difference 7×10^{-8}.
- Uniform scaling test: the target is the source scaled by k about its centre (synthetic plant, 2500 points), so the true $\det J = k^3$.

k	true $\det J = k^3$	median found	5th–95th percentile	$\det J \leq 0$
1.1	1.331	1.254	1.05 to 2.65	0%
1.2	1.728	1.680	1.38 to 2.63	0%
1.5	3.375	3.371	2.91 to 5.30	0%

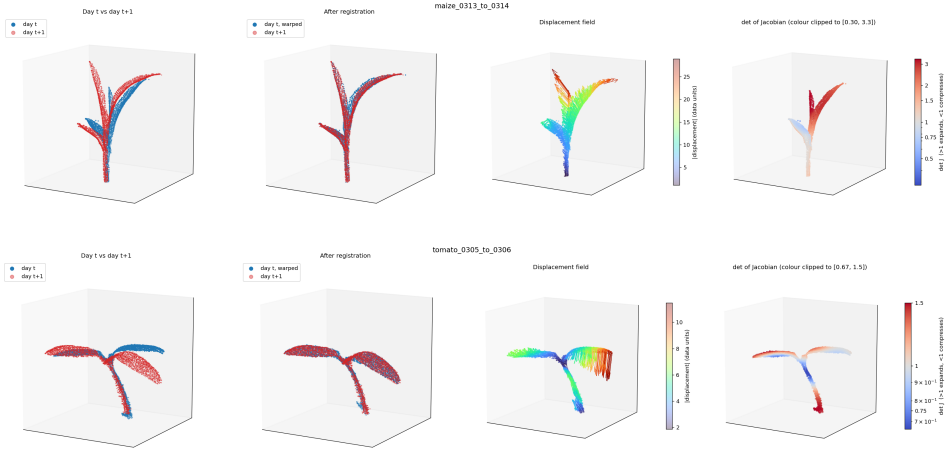
The median tracks the truth. The wide upper tail is local noise at the ends of thin structures, which is why single-point values are not interpreted.

Validation 2: synthetic growing plant



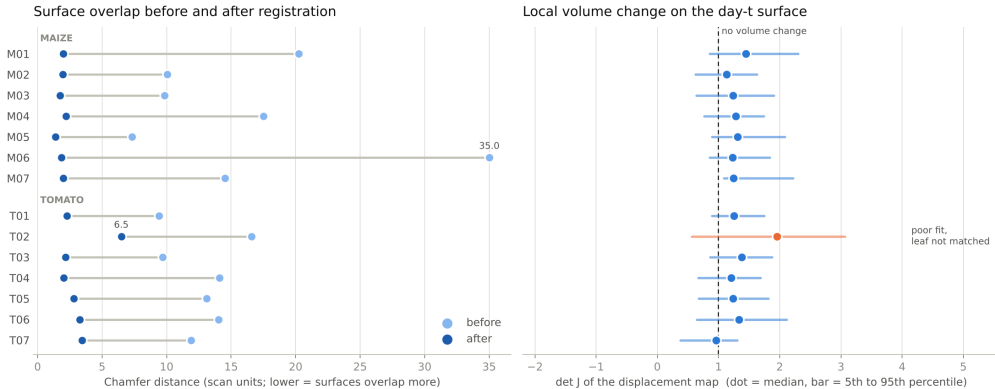
Stem and leaves with known growth; panels as in the real-data plots.

Sanity check on real laser scans (Bonn)



One maize and one tomato pair from PRBonn/4d_plant_registration, one day apart.

Pheno4D: all 14 plants



Left: Chamfer distance before and after. Right: median and 5th–95th percentile of $\det J$. M06 and T02 flagged.

Metrics table (day 0324→0325)

Plant	Chamfer before	Chamfer after	mean $\|u\|$	max $\|u\|$	median det J	det J 5th–95th percentile	det $J \leq 0$ (% of points)
M01	20.2	1.98	28.6	172	1.45	0.86 to 2.30	0.0
M02	10.1	1.97	15.5	58	1.13	0.62 to 1.63	0.0
M03	9.8	1.76	18.4	75	1.24	0.64 to 1.90	0.0
M04	17.5	2.22	23.8	173	1.28	0.76 to 1.74	0.0
M05	7.3	1.37	12.9	44	1.31	0.89 to 2.08	0.0
M06	35.0	1.85	44.8	233	1.23	0.85 to 1.84	0.0
M07	14.5	1.99	20.1	63	1.25	1.09 to 2.22	0.0
T01	9.4	2.29	17.9	41	1.25	0.89 to 1.75	0.0
T02	16.6	6.49	36.9	76	1.96	0.56 to 3.07	2.7
T03	9.7	2.18	19.3	33	1.38	0.86 to 1.87	0.0
T04	14.1	2.01	18.3	43	1.20	0.67 to 1.69	0.0
T05	13.1	2.79	18.9	63	1.23	0.68 to 1.81	0.0
T06	14.0	3.27	26.4	88	1.33	0.65 to 2.11	0.0
T07	11.9	3.44	32.7	61	0.96	0.38 to 1.31	0.0

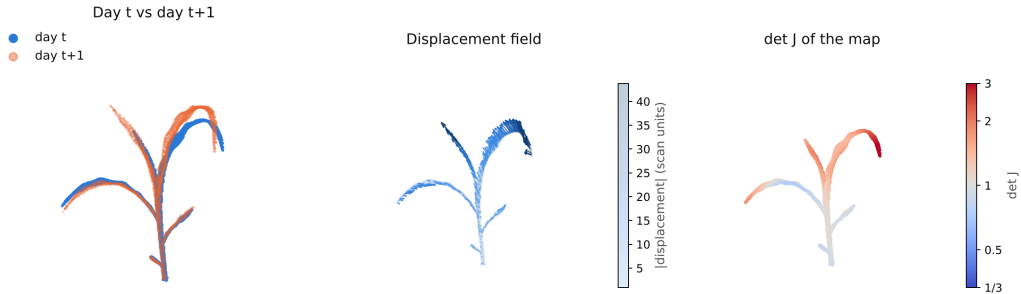
All rows: rigid pre-alignment plus non-rigid CPD. Lengths in scan units (probably mm, unverified).

Default vs rigid pre-alignment, all 14 plants

Plant	Chamfer after		median det J		det $J \leq 0$ (%)	
	default	rigid	default	rigid	default	rigid
M01	1.97	1.98	1.42	1.45	0.0	0.0
M02	1.98	1.97	1.12	1.13	0.0	0.0
M03	1.76	1.76	1.24	1.24	0.0	0.0
M04	2.20	2.22	1.25	1.28	0.0	0.0
M05	1.35	1.37	1.29	1.31	0.0	0.0
M06	2.00	1.85	-1.16	1.23	90.0	0.0
M07	1.99	1.99	1.20	1.25	0.0	0.0
T01	2.29	2.29	1.26	1.25	0.0	0.0
T02	6.73	6.49	1.85	1.96	1.5	2.7
T03	2.18	2.18	1.38	1.38	0.0	0.0
T04	2.01	2.01	1.20	1.20	0.0	0.0
T05	2.79	2.79	1.23	1.23	0.0	0.0
T06	3.28	3.27	1.34	1.33	0.0	0.0
T07	3.44	3.44	0.96	0.96	0.0	0.0

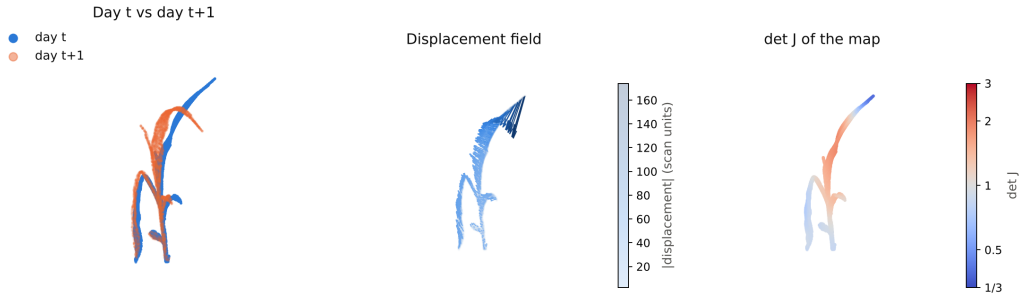
Same parameters in both runs. For 12 plants the two agree to within 0.05 in median det J. M06: the default folds, rigid fixes it. T02: rigid changes little and slightly raises the folding fraction.

M05: the cleanest case



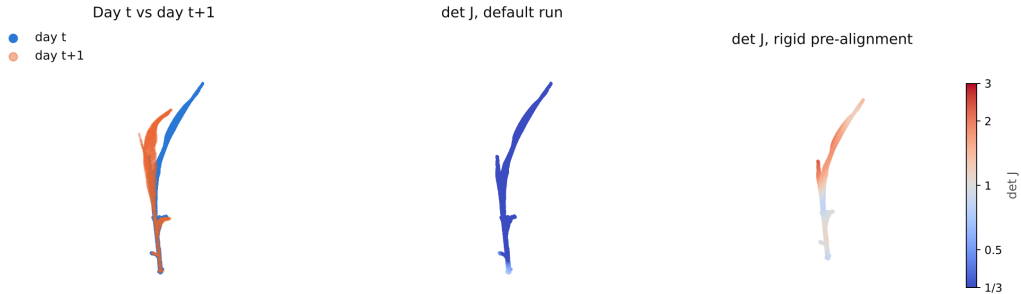
Median det J 1.29, no folding. Expansion concentrates in the upper, youngest leaves; older leaves stay near 1.

M04: a folding leaf tip



The tall tip curls over (displacement up to 173) and reads as compression (blue). This is bending, not shrinking.

M06: the default fit folds, rigid pre-alignment fixes it



Default: 90% of det $J \leq 0$, so the all-blue panel is meaningless (negatives clip to the lowest colour). With rigid pre-alignment: median det J 1.23, 0% negative.

T02: a poor fit



A leaf moved far and has no counterpart; rigid pre-alignment does not rescue it. Do not trust this field.

How to read the plots

- **Overlay** (blue: day t; orange: day $t{+}1$): where the plant moved.
- **Displacement field:** long arrows are mostly leaf motion (drooping, swinging), not growth.
- $\det J$ (log colour scale, 1/3 to 3): red is expansion, blue is compression. Negatives clip to the lowest colour.
- **Bending signature:** compression on one side of a leaf with expansion on the other.
- **Failure signatures:** arrows that tangle, vertical streaks under the plant (an unmatched leaf), a dark patch of values far below 1.

Limitations

- No ground truth on real data; “good” means low Chamfer distance and a plausible field.
- CPD knows nothing about organs. New or lost leaves are handled only by the outlier term, and can still drag nearby points.
- $\det J$ is unreliable near new tissue and where leaves bend or rotate.
- One labelled consecutive pair per plant; 3000-point subsample; parameters tuned on synthetic growth.
- Rigid pre-alignment gives a global shift/rotation only; it cannot fix a leaf that has no counterpart (T02).

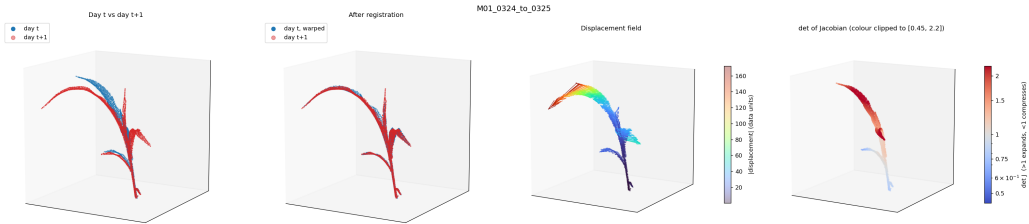
Next steps

- Per-organ registration using the Pheno4D labels; Hungarian matching for unlabelled scans.
- Symmetric outlier handling with explicit new-organ and lost-organ masks.
- Areal stretch from surface normals instead of volumetric $\det J$.
- Longer baselines (gap of two days) and a sweep over β, λ, w per plant.

References

- Myronenko & Song, Point set registration: Coherent Point Drift, IEEE TPAMI 2010.
- Schunck et al., Pheno4D: a spatio-temporal dataset of maize and tomato plant point clouds, PLOS ONE 16(8):e0256340, 2021.
- Chebrolu et al., Registration of spatio-temporal point clouds of plants for phenotyping, ICRA 2020.
- Magistri et al., Segmentation-based 4D registration of plants point clouds, IROS 2020.
- Li et al., Analyzing growing plants from 4D point cloud data, SIGGRAPH Asia 2013.

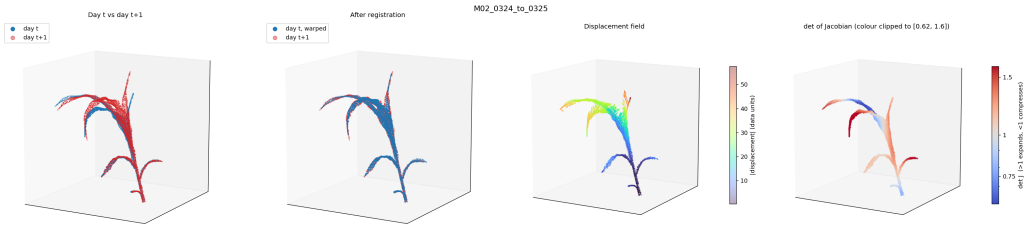
M01 (maize): day 0324 → 0325



Rigid pre-alignment. Chamfer 20.2 → 1.98 mean $|u|$ 28.6 max $|u|$ 172 median det J 1.45 det $J \leq 0$: 0.0%

One large arching leaf swings up and outward (max displacement 172). Its tip expands strongly (det J about 2 or more); the small lower leaf stays near 1.

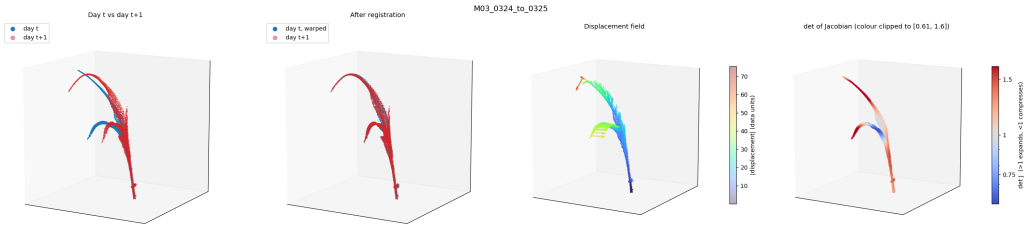
M02 (maize): day 0324 → 0325



Rigid pre-alignment. Chamfer 10.1 → 1.97 mean $|u|$ 15.5 max $|u|$ 58 median det J 1.13 det $J \leq 0$: 0.0%

Upper leaves and the central leaf tip move most (up to 58). Det J alternates blue and red along the arching leaf: bending and sliding, not growth.

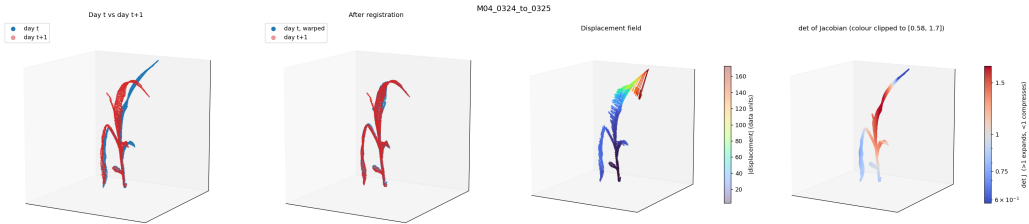
M03 (maize): day 0324 → 0325



Rigid pre-alignment. Chamfer 9.8 → 1.76 mean $|u|$ 18.4 max $|u|$ 75 median det J 1.24 det $J \leq 0$: 0.0%

The big leaf droops at its tip and the tip expands. The smaller leaf reads blue and moves sideways: motion misread as compression.

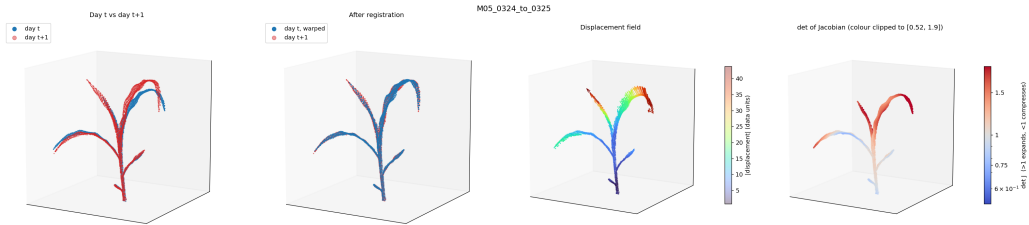
M04 (maize): day 0324 → 0325



Rigid pre-alignment. Chamfer 17.5 → 2.22 mean $|u|$ 23.8 max $|u|$ 173 median det J 1.28 det $J \leq 0$: 0.0%

The tall leaf tip curls over (arrows up to 173) and reads blue (below 0.6): a folding leaf, not a shrinking one. Treat as an artefact.

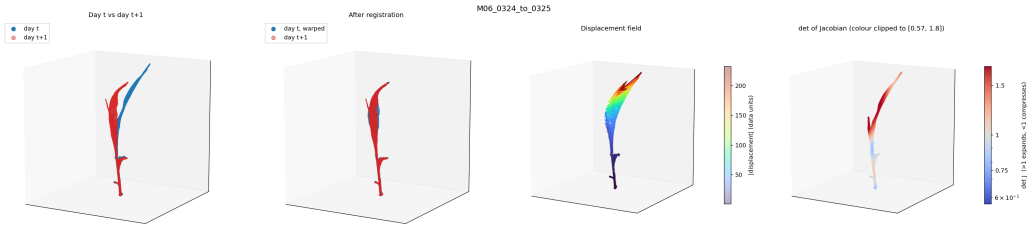
M05 (maize): day 0324 → 0325



Rigid pre-alignment. Chamfer 7.3 → 1.37 mean $|u|$ 12.9 max $|u|$ 44 median det J 1.31 det $J \leq 0$: 0.0%

Cleanest maize result. Det J is 1.5 to 1.9 on the upper right leaf and near 1 on older leaves: growth concentrated in the youngest leaves.

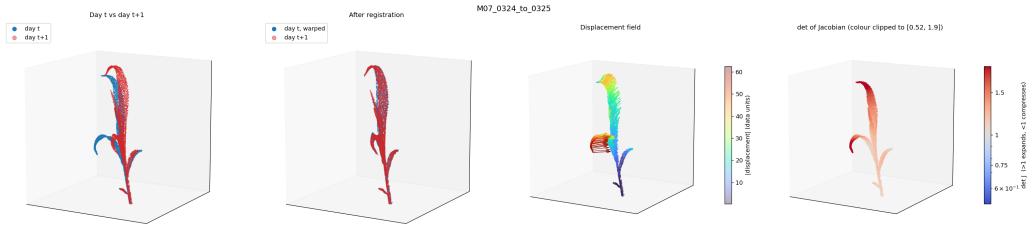
M06 (maize): day 0324 → 0325



Rigid pre-alignment. Chamfer 35.0 → 1.85 mean $|u|$ 44.8 max $|u|$ 233 median det J 1.23 det $J \leq 0$: 0.0%

Largest overall motion (max 233). The default run folded here (90% negative det J); with rigid pre-alignment, median det J 1.23 and 0% negative. Expansion sits in the upper leaf.

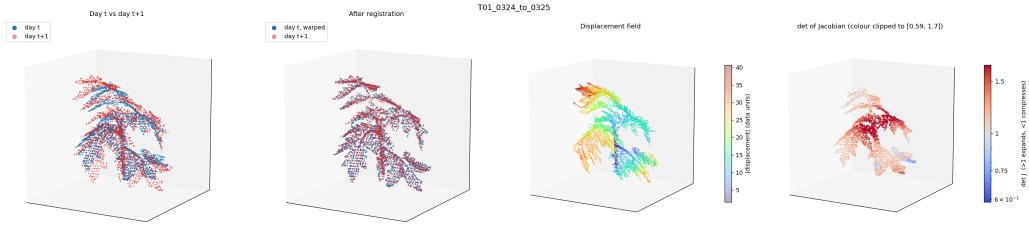
M07 (maize): day 0324 → 0325



Rigid pre-alignment. Chamfer 14.5 → 1.99 mean $|u|$ 20.1 max $|u|$ 63 median det J 1.25 det $J \leq 0$: 0.0%

Lower-left leaf swings out sideways (about 60) and the central leaf top expands. 5th percentile 1.04: no compression anywhere. A thin new leaf at the top is left unmatched.

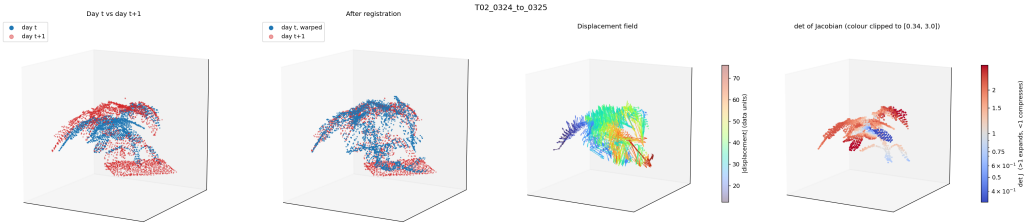
T01 (tomato): day 0324 → 0325



Rigid pre-alignment. Chamfer 9.4 → 2.29 mean $|u|$ 17.9 max $|u|$ 41 median det J 1.25 det $J \leq 0$: 0.0%

Decent fit. The central leaf cluster expands (1.5 or more); the outer leaves sit near 1.

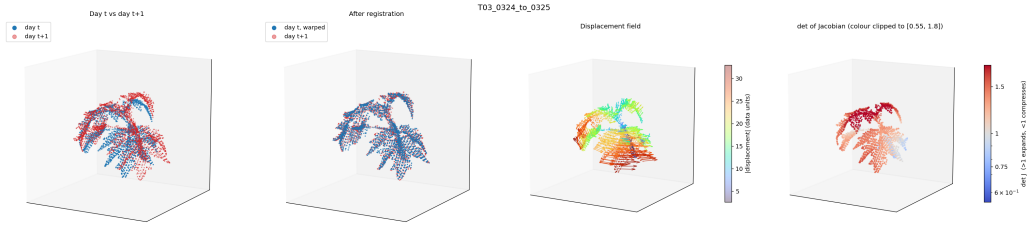
T02 (tomato): day 0324 → 0325



Rigid pre-alignment. Chamfer 16.6 → 6.49 mean $|u|$ 36.9 max $|u|$ 76 median det J 1.96 det $J \leq 0$: 2.7%

Worst fit; rigid pre-alignment does not help (2.7% negative det J, slightly worse than the default 1.5%). A leaf dropped far below its day-t position and is unmatched; arrows tangle and det J has extreme patches. Do not trust.

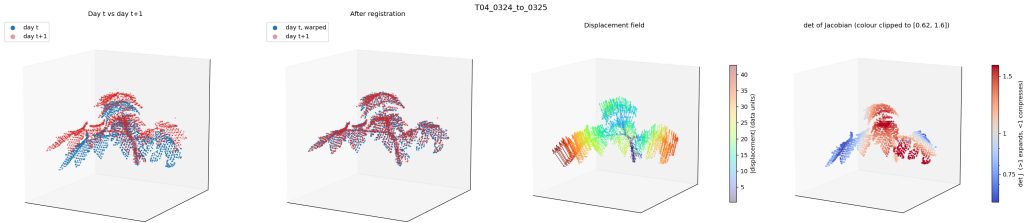
T03 (tomato): day 0324 → 0325



Rigid pre-alignment. Chamfer 9.7 → 2.18 mean $|u|$ 19.3 max $|u|$ 33 median det J 1.38 det $J \leq 0$: 0.0%

Lower leaves swing outward (about 30); the upper arc leaf expands (above 1.5). Reasonable fit.

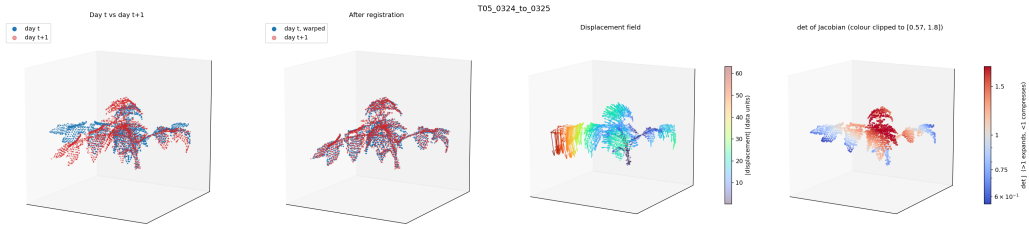
T04 (tomato): day 0324 → 0325



Rigid pre-alignment. Chamfer 14.1 → 2.01 mean $|u|$ 18.3 max $|u|$ 43 median det J 1.20 det $J \leq 0$: 0.0%

Outer leaves droop downward (about 40). The blue left leaf is probably motion misread as compression.

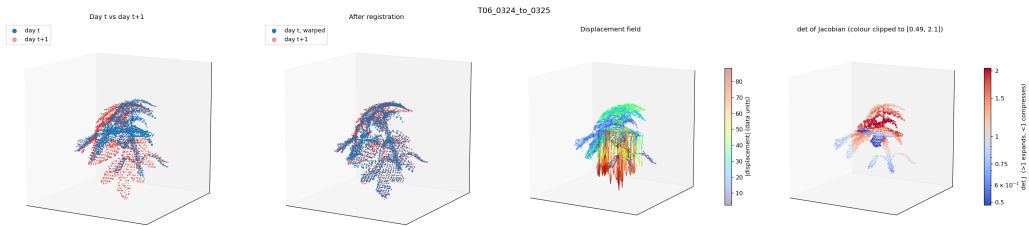
T05 (tomato): day 0324 → 0325



Rigid pre-alignment. Chamfer 13.1 → 2.79 mean $|u|$ 18.9 max $|u|$ 63 median det J 1.23 det $J \leq 0$: 0.0%

Left leaf droops by up to about 60. The central growing tip expands (above 1.5); the blue outer leaves are probably artefacts.

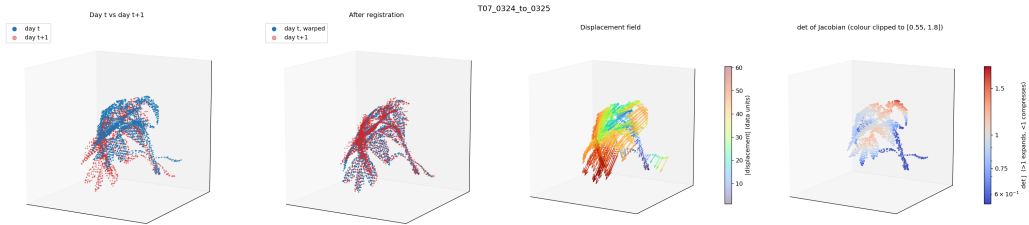
T06 (tomato): day 0324 → 0325



Rigid pre-alignment. Chamfer 14.0 → 3.27 mean $|u|$ 26.4 max $|u|$ 88 median det J 1.33 det $J \leq 0$: 0.0%

Vertical streaks of up to about 88 under the plant: a lower leaf at day $t + 1$ is unmatched, and the dark blue central blob (about 0.5) is points pulled onto it. Do not trust.

T07 (tomato): day 0324 → 0325



Rigid pre-alignment. Chamfer 11.9 → 3.44 mean $|u|$ 32.7 max $|u|$ 61 median det J 0.96 det $J \leq 0$: 0.0%

Poor fit. A long thin leaf at the right has no counterpart at day $t+1$, and much of the plant reads 0.4 to 0.7. Only plant with median det J below 1; do not read as shrinkage.